Introduction

Goal of the Study

We are going to evaluate the system “StatePX” and its capabilities. We are not evaluating you personally in any way.
The collected data is pseudo-anonymous.
It might be possible that tasks are not manageable in the given time frame. Again, this is no evaluation of your skill. The tasks are designed in a way to gauge how far users can get, so they are intentionally extensive.

General Introduction

StatePX was developed to model “player experiences”, an experience a player has while playing a game.
The experience is modeled with a graph, where a node describes a player experience and edges describe how these experiences are connected.
In addition to just modeling the player experience, it can also analyze the graph, show how the progression unfolds, and give insights into solvability, i.e. whether certain areas can be reached and how.

Each task has a maximum time limit. Work as accurately as you normally would. If you finish before the time limit, let the moderator know. Otherwise, the moderator will stop the task once the time limit has been reached.

Task 1 - Building a first graph

In this task, you get familiar with the tool and build a first graph together with relevant components, locks, and keys. Along with this task you can see the final section of a dungeon from Legend of Zelda.

We have already prepared a node for the first room, but it is still missing many details. For all sub-tasks, include all information that is represented accompanying facts of the map. During all tasks, feel free to ask any clarifying questions regarding the tasks.

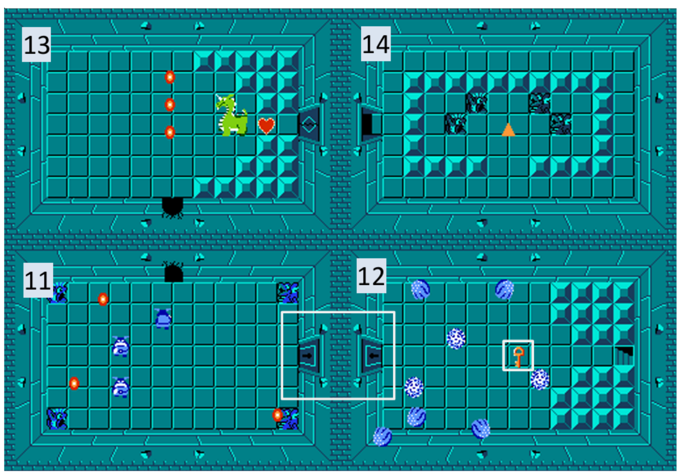
Task 1a

Complete the node for room 12 by adding all components and all keys.

Task 1b

Complete the graph by adding all nodes, edges, components, locks, and keys for rooms 11, 12, 13, and 14.

Facts:
The final boss of the dungeon. Enemies: 1 Boss Name: "Aquamentus"
The room before the boss. Enemies: 3



Facts:
The final room of the dungeon. Triforce
The first room of the final floor. Enemies: 8 Simple Key: 1

Clarifying hints:

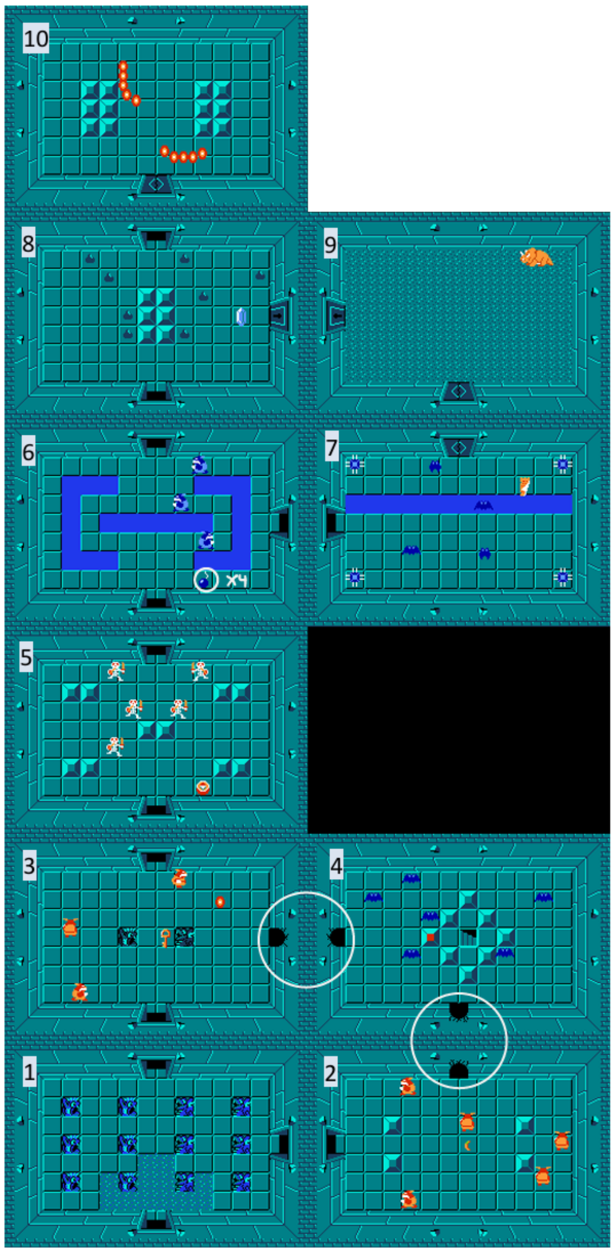
- Edges can be uni- or bi-directional.
- Keys and locked doors are marked by boxes

We just provided you with another section of the dungeon that leads to the final section from task 1. The two parts of the dungeon are connected through staircases in room 4 and room 12. Similar to task 1a, we prepared part of the graph for this dungeon, but it is still missing some nodes, locks, keys and components.

Task 1c

Complete the graph for the dungeon by adding all nodes, edges, components, locks, and keys for rooms 5, 6, and 7.

Facts:
A dead end where the player has to fight.
-
This has water where the player can not walk. Enemies: 3 Bombs: 4
A room where the player can find a compass. Enemies: 5
-
-



Facts:
-
This room is divided into two parts by water. Enemies: 8
-
-

Gameplay Analysis

In this part of the study, you will use dungeon graphs to analyze different aspects of gameplay design.

You will complete two different analysis tasks using both the tool and a virtual whiteboard (Miro). Depending on the study condition, the order of tools and tasks may differ.

Throughout the tasks, the following progression elements are used:

- Simple Keys unlock Simple Doors and Wooden Doors
- Boss Keys unlock Boss Doors
- Bombs unlock Bombable Walls and Wooden Doors
- Abilities unlock Ability Locks

Task 2 - Solvability Analysis

This task focuses on analyzing the solvability of another Zelda dungeon.

During gameplay players collect keys, abilities, and other progression items to unlock new areas of the dungeon. Incorrect placement of these progression elements can make parts of the dungeon unreachable or even impossible to complete.

Your goal is to analyze the given graph and identify any potential solvability issues.

These may include, but are not limited to:

- Missing keys (Simple Key, Boss Key, Bombs, Abilities)
- Unreachable Sections
- Soft-locks (situations where player may or may not be able to continue)
- Hard-locks (situations where the player is not able to continue)

Miro

This virtual whiteboard models the dungeon as a graph.

The symbols shown on the edges indicate which progression item is required to traverse that connection.

Use the graph to identify any potential solvability issues.

You are allowed to add sticky notes of a different color after identifying and communicating an issue. Also let us know your confidence regarding the issue.

StatePX

StatePX provides a pathfinding function that can determine whether a valid path exists between rooms.

To calculate a path:

- Click the starting room
- Hold **Ctrl** and **click** the destination room
- Wait a moment to let the tool calculate the path
- Repeat **Ctrl+Click** to add additional destinations and chain multiple paths together.

Use both the graph and the pathfinding functionality to identify any potential solvability issues.

You are allowed to make modifications to the graph after identifying and communicating an issue. Also let us know your confidence regarding the issue.

A legend describing the visualization colors is provided next to the interface.

Task 3 - Pacing Analysis

This task focuses on analyzing the pacing of combat encounters throughout the dungeon.

Each room contains information about

- the number of enemies
- the time required to defeat all enemies (room completion time)

The dungeon can be divided into three gameplay sections:

- **Section A:** from **START** to the room where the player receives a **New Ability**
- **Section B:** from the **New Ability** room to the room containing the **Boss Key**
- **Section C:** from the **Boss Key** room to the **BOSS**

Answer the following questions.

1. Is the number of enemies steadily increasing throughout the dungeon?
2. Does it take longer on average to complete rooms in Section A than in Section B?
3. Are there more enemies in Section C than in Section B?
4. Which room most likely contains a miniboss?
5. Between which two consecutive rooms does the number increase the most?
6. Which room after a high-intensity encounter has notably fewer enemies or lower completion time?

Miro

Use the information shown in the graph to answer the questions.

When reasoning about the dungeon, make sure to consider valid progression paths through the dungeon.

StatePX

In addition to the pathfinding functionality used in the previous task, StatePX provides diagrams for visualizing numerical information.

To create and use a diagram:

- Click the + button in the lower-left corner.
- Select one or more numerical components as f(x).
- Select a path like before.

The diagram visualizes the selected values across the dungeon graph.

Whenever a path is calculated or reset, the diagram automatically updates to visualize the selected values along that path.

Use the graph, the pathfinding functionality, and the diagrams to answer the questions.

For us

- fix softlock not showing
- Only help with task clarification, never with interface clarification
- We allow modifications in pix:e as it is the intended usage. For miro, we allow them to add new notes in a specific color so we can easily delete them later

Using SUS

The SU scale is generally used after the respondent has had an opportunity to use the system being evaluated, but before any debriefing or discussion takes place. Respondents should be asked to record their immediate response to each item, rather than thinking about items for a long time.
All items should be checked. If a respondent feels that they cannot respond to a particular item, they should mark the centre point of the scale.

Metrics

Part 1:

- Interface Help Requests
- Error Rate / Undo / Redo
- Time to finish 1a, 1b, 1c
- Completion rate 1a, 1b, 1c
- Accuracy of completion (not necessarily)

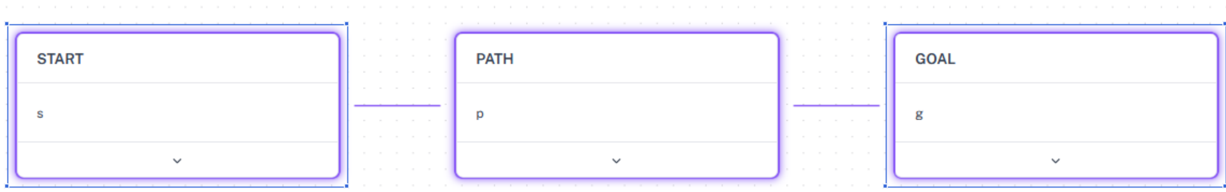
Part 2:

- Time per issue
- Quality of explanation
- Confidence
- Number of issues (tracked by tracking issues)
- Distinct issue types (tracked by tracking issues)
- Time to finish before time is over (tracked by tracking time per issue)

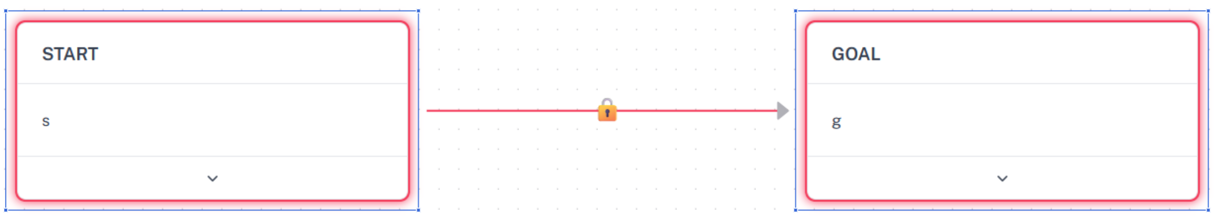
User Groups

	Miro First	StatePx First
A first	P1	
B first		

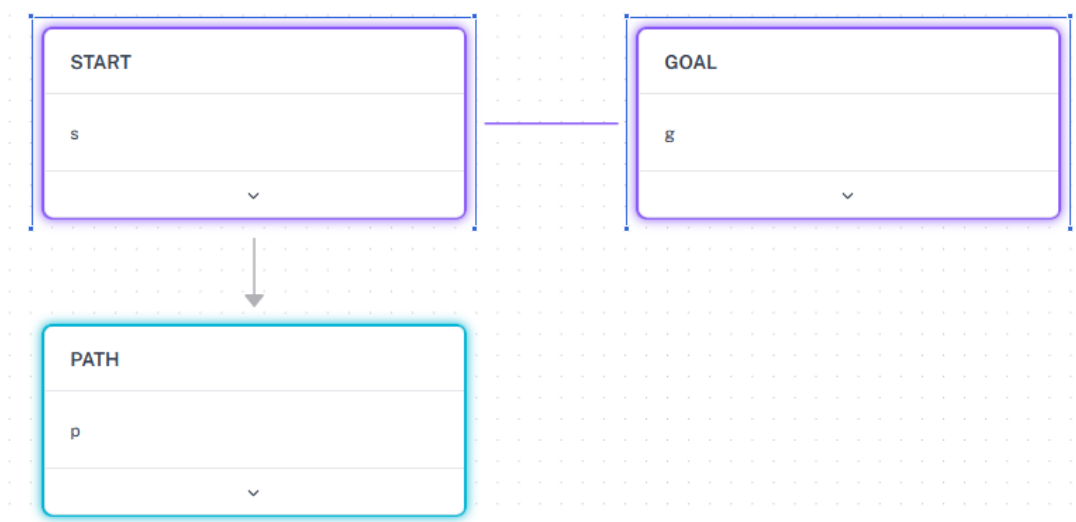
Legend



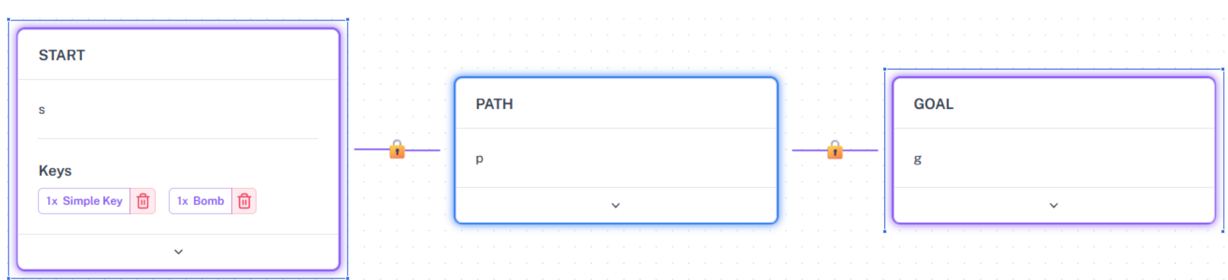
Purple: Nodes and edges along a valid path from START to GOAL are highlighted in purple.



Red: Selected nodes are highlighted in red if there is no valid path. Edges that could not be unlocked are highlighted as well.



Light Blue: Nodes are highlighted in light blue if there are no outgoing edges.



Dark Blue: Nodes that contain a potential to softlock are highlighted in dark blue. A potential to softlock occurs when a lock can be opened by more than one key type.

Task 1 - Open Ended Question

Was there anything about the modeling interface that you found confusing or frustrating?

Task 2+3 - Comparison to Miro

	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
Compared to Miro, StatePX made it easier to analyze dungeon designs.					
Compared to Miro, I felt more confident in my answers.					
Compared to Miro, I was able to complete the tasks more efficiently.					
If I were analyzing game levels in practice, I would prefer using StatePX over a general-purpose whiteboard.					